Talentree

Software Requirements Specification (SRS)

Version: 1.0
Date: October 2025
Project Team:

- Backend Team: ASP.NET Core
- Frontend Team: Angular
- AI & Data Science Team: Python Microservices

Document Control

Version	Date	Author	Changes
1.0	October 2025	Talentree Team	Initial release

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document provides a comprehensive description of the Talentree platform. It details the functional and non-functional requirements for the Minimum Viable Product (MVP) that will be developed by the project team.
The intended audience for this document includes:

- Development teams (Backend, Frontend, AI/Data Science)
- Project stakeholders and advisors
- Quality assurance and testing teams
- Future maintenance and enhancement teams

1.2 Project Scope

Talentree is a digital marketplace platform designed to empower small businesses and individual entrepreneurs in the creative industries. The platform serves three distinct user groups and provides the following core capabilities:

In Scope (MVP):

- Multi-vendor marketplace for handmade and creative products
- Raw material sourcing for business owners
- Order management and transaction processing
- AI-powered product and material recommendations
- Business analytics and insights
- Admin management tools for platform operations
- Secure payment processing via Stripe
- Real-time notifications system

Out of Scope (Phase 2):

- Direct messaging between buyers and sellers
- Native mobile applications (iOS/Android)
- Multi-language support
- Local Egyptian payment gateways (Paymob, Fawry)
- Loyalty and referral programs
- Advanced predictive analytics

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
BO	Business Owner - A seller on the platform

Admin	Platform administrator with elevated privileges
SRS	Software Requirements Specification
MVP	Minimum Viable Product
API	Application Programming Interface
UI/UX	User Interface / User Experience
EF Core	Entity Framework Core
SQL	Structured Query Language
CRUD	Create, Read, Update, Delete operations
KPI	Key Performance Indicator
2FA	Two-Factor Authentication
SignalR	Real-time web functionality library for ASP.NET

1.4 Document Conventions

Requirement Identifiers

Requirements are identified using the following format:

- FR-XX-YY = Functional Requirement (Module XX, Requirement YY)
- NFR-XX = Non-Functional Requirement (Number XX)

- TR-XX = Technical Requirement (Number XX)

Priority Levels

- Critical: Must be implemented for MVP launch
- High: Should be implemented for MVP launch
- Medium: Nice to have in MVP, required in Phase 2
- Low: Future enhancement

Formatting Conventions

- Bold text indicates important terms or emphasis
- `Code snippets` are shown in monospace font
- [Placeholders] indicate items requiring team input

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2. System Overview

2.1 Product Perspective

Talentree operates as a standalone web-based platform accessible through modern web browsers. The system consists of three main subsystems:

1. Web Application (Frontend): Angular-based single-page application providing responsive UI for all user types
2. RESTful API Backend: ASP.NET Core Web API handling business logic, data persistence, and orchestration
3. AI Microservices: Python-based services providing recommendations, insights, and intelligent assistance

System Context Diagram

```
┌──────────────┐
│   Customers  │
└──────────────┘
       │
       ├──────────────┐
       │              │
┌──────▼──────┐   ┌───▼──────────┐
│  Angular    │   │   Business   │
│  Frontend   │   │   Owners     │
└─────────────┘   └──────────────┘
       │              │
       └──────────────┘
              │
              │
```

```
        ┌──────────▼──────────┐
        │   ASP.NET Core      │
        │    Web API          │
        └─────────────────────┘
                  │
        ┌─────────┴──────────────────────────────────────┐
        │                   │                   │                  │
┌───▼────┐   ┌──────────▼──────┐   ┌───▼────────┐   ┌───▼──────────┐
│ SQL    │   │ Python AI       │   │ Stripe     │   │ SignalR      │
│ Server │   │ Services        │   │ Payment    │   │ Hub          │
└────────┘   └─────────────────┘   └────────────┘   └──────────────┘
```

2.2 Product Features (High-Level)

2.2.1 Business Owner Features

- Product catalog management with admin approval workflow
- Order tracking and fulfillment management
- Raw material procurement from verified vendors
- Financial reporting and insights
- Customer feedback management
- AI-powered business analytics
- Support ticket system

2.2.2 Customer Features

- Product browsing and search
- Brand discovery and exploration
- Shopping cart and checkout
- Wishlist/favorites management
- Order tracking
- Product reviews and ratings
- Personalized recommendations

2.2.3 Admin Features

- User management (sellers and buyers)
- Product approval and moderation
- Order and transaction oversight
- Platform analytics and reporting
- Vendor management
- Educational content management
- System configuration and settings
- Support and complaint resolution

2.2.4 AI-Powered Features

- Product recommendation engine
- Raw material recommendation system
- Business insights dashboard
- Admin insights dashboard
- Smart pricing suggestions
- AI business assistant chatbot
- Marketing plan generator

2.3 User Classes and Characteristics

Customer

- Technical Expertise: Low to Medium
- Primary Activities: Browse, search, purchase products
- Access Frequency: Occasional to frequent
- Privileges: View public content, manage own orders and wishlist

Business Owner

- Technical Expertise: Medium
- Primary Activities: Manage inventory, fulfill orders, analyze sales
- Access Frequency: Daily
- Privileges: Manage own products, orders, and business profile

Administrator

- Technical Expertise: High
- Primary Activities: Platform management, user support, analytics review
- Access Frequency: Daily
- Privileges: Full system access, user management, platform configuration

System (AI Services)

- Type: Automated service
- Primary Activities: Generate recommendations, analyze data, provide insights , assistant chatbot
- Access Frequency: Continuous
- Integration: API-based communication with backend

2.4 Operating Environment

Client-Side Requirements

- Supported Browsers: Chrome 90+, Firefox 88+, Safari 14+, Edge 90+

- Screen Resolutions: 320px (mobile) to 2560px (desktop)
- Internet Connection: Minimum 1 Mbps for basic functionality

Server-Side Environment

- Operating System: Windows Server 2019+ or Linux (Ubuntu 20.04+)
- Runtime: .NET 8.0 as (LTS)
- Database: Microsoft SQL Server 2019+
- Python Environment: Python 3.9+ for AI services
- Hosting: Development + Demo → Render Hosting (free & simple) , For Production Launch → Migrate to Azure Cloud

2.5 Design and Implementation Constraints

Technical Constraints

- Must use ASP.NET Core for backend API development
- Frontend must be developed using Angular framework
- Database must be Microsoft SQL Server
- AI services must be Python-based and API-accessible
- Payment processing limited to Stripe in MVP
- Real-time features must use SignalR

Business Constraints

- MVP must support only three business categories initially
- Arabic language support deferred to Phase 2
- Mobile apps deferred to Phase 2
- The platform must comply with Egyptian e-commerce and data protection regulations, ensuring transparent pricing, user data privacy, and secure online payment handling.

Security Constraints

- All data transmission must use HTTPS
- User passwords must be hashed using industry-standard algorithms
- Payment information must never be stored on platform servers

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3. MVP Scope & Business Categories

3.1 MVP Goals & Objectives

The Talentree MVP aims to validate the core marketplace concept while delivering immediate value to early adopters. The primary objectives are:

Primary Goals

1. **Empower small businesses** to establish and manage their online presence through accessible digital and e-commerce tools.
2. **Streamline raw material sourcing** by connecting business owners with verified and reliable suppliers.
3. **Leverage AI** to deliver personalized recommendations, enhancing product discovery and overall user experience.
4. **Maintain platform integrity and safety** through comprehensive admin tools for content moderation and user management.
5. **Generate actionable business insights** using data-driven analytics to support informed decision-making and sustainable growth.

Success Metrics (MVP) - After Hosting

- Successfully onboard 50+ verified business owners
- Achieve 500+ registered customers
- Process 200+ completed transactions
- Maintain 95%+ platform uptime
- Achieve 70%+ user satisfaction rating

3.2 Supported Business Categories

To ensure focused growth and effective supplier relationships, the MVP supports three carefully selected business categories:

3.2.1 Fashion & Accessories

Description: Clothing, jewelry, bags, and fashion accessories crafted by independent designers and small manufacturers.
Target Sellers: Fashion designers, tailors, jewelry makers, leather craftsmen
Primary Raw Materials:

- Fabrics (cotton, linen, denim, silk)
- Sewing threads and notions
- Zippers, buttons, and fasteners
- Leather and faux leather materials
- Beads, stones, and jewelry findings
- Packaging materials (boxes, bags, tags)

Example Products: Handmade dresses, custom jewelry, leather bags, embroidered accessories

3.2.2 Handmade Crafts

Description: Art pieces, home décor, and functional craft items created through traditional or modern crafting techniques.
Target Sellers: Artisans, potters, woodworkers, resin artists, candle makers
Primary Raw Materials:

- Wax (soy, beeswax, paraffin)
- Resin and hardeners
- Clay and pottery materials
- Yarn and textile fibers
- Natural dyes and pigments
- Decorative embellishments
- Molds and casting materials
- Packaging (boxes, wrapping paper)

Example Products: Scented candles, resin art, pottery, macramé décor, wooden crafts

3.2.3 Natural & Beauty Products

Description: Organic skincare, haircare, and wellness products made from natural ingredients.
Target Sellers: Organic product makers, aromatherapists, soap makers, herbalists
Primary Raw Materials:

- Essential oils and fragrance oils
- Carrier oils (coconut, jojoba, argan)
- Shea butter and cocoa butter
- Soap bases (melt-and-pour, cold process)
- Natural preservatives
- Botanical extracts and herbs
- Glass jars and bottles
- Pumps, droppers, and applicators
- Labels and product tags
- Natural colorants (clays, micas)

Example Products: Handmade soaps, body butters, essential oil blends, natural face serums

3.3 Raw Materials Mapping

Each business category is associated with specific suppliers and material types to streamline procurement:

Business Category	Material Categories	Supplier Types
Fashion & Accessories	Textiles, Notions(buttons ,zippers ..), Hardware	Fabric wholesalers, Thread suppliers,

		Packaging vendors
Handmade Crafts	Art Supplies, Casting Materials	Craft stores, Specialty suppliers, Resin distributors
Natural & Beauty	Ingredients, Packaging, Labels	Organic suppliers, Glass/bottle vendors, Print shops

Benefits of Focused Categories:

- Simplified vendor management for admins
- Better AI recommendations due to focused data
- Clearer brand positioning
- Stronger community and networking opportunities
- Easier quality control and compliance

3.4 Phase 2 Roadmap (Future Features)

Planned Enhancements (Post-MVP)

Marketplace Expansion:

- Add 3-5 additional business categories
- Multi-vendor promotion tools
- Bulk order management
- Seasonal campaign features

Communication:

- Full messaging system between buyers and sellers
- Video call support for custom orders
- Seller shop announcements

Advanced Analytics:

- Predictive sales forecasting
- Customer lifetime value analysis

- Inventory optimization recommendations
- Competitive pricing intelligence

Monetization & Growth:

- Loyalty points and rewards program
- Referral system with incentives
- Premium seller subscriptions
- Featured product placements

Localization:

- Arabic language interface
- Local payment gateway integration (Paymob, Fawry)
- Regional shipping partners
- Currency localization (EGP support)

Mobile Applications:

- Native iOS app
- Native Android app
- Push notifications
- Offline mode for browsing

AI Enhancements:

- Voice-activated assistant
- Automated customer service chatbot
- Fraud detection system
    - Those depend on “Data maturity and sufficient user activity for model training.”

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4. User Roles & Permissions

4.1 Role Definitions

4.1.1 Customer (Buyer)

Description: End users who browse and purchase products from business owners on the platform.

Primary Responsibilities:

- Browse and search products
- Purchase items
- Review and rate products
- Manage personal profile and orders

System Access: Public marketplace and personal account areas

4.1.2 Business Owner (Seller)

Description: Small business owners and entrepreneurs who list and sell products, and source raw materials.
Primary Responsibilities:

- Manage product catalog
- Fulfill nd track customer orders
- Purchase raw materials
- Monitor business performance
- Respond to customer reviews
- Access educational materials to enhance business skills
- Interact with intelligent assistant chatbots for guidance, support

System Access: Business dashboard and marketplace (as buyer for raw materials)

4.1.3 Administrator

Description: Platform operators responsible for maintaining quality, resolving issues, and managing system configuration.
Primary Responsibilities:

- Review and approve/reject seller registrations with optional automatic approval after a defined time period (e.g., 12 hours) if no manual action is taken by the admin.
- Moderate and verify product listings to ensure compliance with platform policies, content guidelines, and quality standards before publish
- Handle disputes, complaints, and policy violations, ensuring fair resolution and user accountability.
- Manage and validate vendor and supplier profiles, including verification and ongoing performance evaluation.
- Configure and update platform settings, including pricing rules, categories, and service configurations.
- Monitor platform performance, uptime, and usage metrics to maintain operational stability and detect technical or business anomalies.

System Access: Full system access via admin panel

4.1.4 Super Administrator

Description: The Super Administrator holds the highest privilege level within the platform, with unrestricted access to all administrative functions, user role management, and security configurations.

Primary Responsibilities:

- Create, edit, and deactivate administrator accounts.
- Define and configure role-based access control (RBAC) policies.
- Approve and oversee critical system configuration or deployment changes.
- Access and review audit logs, security alerts, and permission histories.
- Ensure compliance with platform security and operational standards.

System Access:
Full access to all system modules and administrative controls, including admin management, configuration settings, and system security operations.

4.2 Permission Matrix

Feature/Action	Customer	Business Owner	Administrator	Super Admin
Authentication				
Register Account	✓	✓	-	-
Login/Logout	✓	✓	✓	✓
Reset Password	✓	✓	✓	✓
Enable 2FA	✓	✓	✓	✓
Product Browsing				
View Products	✓	✓	✓	✓
Search Products	✓	✓	✓	✓
View Brand Pages	✓	✓	✓	✓
Shopping				

Add to Cart	✓	✓ (materials)	-	-
Checkout	✓	✓ (materials)	-	-
Add to Wishlist	✓	-	-	-
Product Management				
Create Product	-	✓	-	-
Edit Own Product	-	✓	-	-
Delete Own Product	-	✓	-	-
Approve Product	-	-	✓	✓
Reject Product	-	-	✓	✓
Delete Any Product	-	-	✓	✓
Order Management				
Place Order	✓	✓ (materials)	-	-

View Own Orders	✓	✓	-	-
Cancel Own Order	✓	-	-	-
Update Order Status	-	✓	✓	✓
Process Refund	-	✓	✓	✓
View All Orders	-	-	✓	✓
Reviews & Ratings				
Write Review	✓	-	-	-
Respond to Review	-	✓	-	-
Delete Review	-	-	✓	✓
User Management				
Edit Own Profile	✓	✓	✓	✓
View User List	-	-	✓	✓
Block User	-	-	✓	✓

Approve Seller	-	-	✓	✓
Delete User	-	-	-	✓
Financial				
View Own Finances	-	✓	-	-
Request Payout	-	✓	-	-
View Platform Revenue	-	-	✓	✓
Process Payouts	-	-	✓	✓
Content Management				
Submit Support Ticket	✓	✓	-	-
Respond to Ticket	-	-	✓	✓
Manage Vendors	-	-	✓	✓
Upload Educational Content	-	-	✓	✓

Platform Settings				
View Settings	-	-	✓	✓
Edit Settings	-	-	-	✓
Manage Admin Accounts	-	-	-	✓
Access Audit Logs	-	-	-	✓

Legend:

- ✓ = Permitted
- = Not Permitted

4.3 Access Control Rules

Rule 1: Data Isolation

Users can only access their own data unless explicitly granted broader permissions. Business owners see only their own products, orders, and financial data. Customers see only their own orders and personal information.

Rule 2: Approval Workflow

New business owner accounts must be approved by an administrator before gaining full access to seller features. Products submitted by business owners require admin approval before appearing in the marketplace.

Rule 3: Action Logging

All administrative actions affecting other users (blocks, deletions, approvals) must be logged with timestamp, admin identifier, and reason.

Rule 4: Cascade Restrictions

When a business owner account is blocked or suspended, all their products are automatically hidden from the marketplace. Active orders remain visible for fulfillment obligations.

Rule 5: Session Management

User sessions must expire after 24 hours of inactivity. Administrative sessions must expire after 8 hours of inactivity. Users must re-authenticate for sensitive operations (profile changes, payment methods).

Rule 6: Restricted Operations

Certain operations require elevated confirmation:

- Account deletion requires password re-entry
- Payout requests require 2FA verification (if enabled)
- Administrative account creation requires Super Admin approval

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5. Functional Requirements

5.1 Business Owner Module

5.1.1 Registration & Onboarding

FR-BO-01: Business Owner Registration Form

- Priority: Critical
- Description: The system shall provide a registration form for new business owners to create an account.
- Inputs:
    - **Personal Information:** Full name, email, phone number, passwor
    - **Business Information:** Business name, business type (from predefined categories), business description, business address *(optional for MVP)*
    - **Legal Information:** Tax ID / Commercial Registration *(optional for MVP)*
    - **Optional Links:** Social media profiles (Facebook, Instagram, or website)
- Processing:
    - Validate all required fields
    - Check for duplicate email/phone
    - Hash and store password securely
    - Set account status to "Pending Approval"
- Outputs:
    - Account created with Pending status
    - Verification email sent to provided address
- Acceptance Criteria:
    - Form validates input in real-time
    - Password must meet strength requirements (8+ characters, mixed case, numbers)
    - Business name must be unique across platform

FR-BO-02: Email Verification

- Priority: Critical
- Description: New business owners must verify their email address before the account can be reviewed by admin.
- Process:
    - System sends verification email with unique token
    - Token expires after 24 hours
    - User clicks link to verify email
    - Account status updates to allow admin review
- Acceptance Criteria:
    - Verification email sent within 1 minute of registration
    - Invalid or expired tokens show appropriate error message
    - User can request new verification email

FR-BO-03: Admin Approval Workflow

- Priority: Critical
- Description: Administrators review and approve/reject business owner applications.
- **Process:**
    - Admin receives notification of pending application.
    - Admin reviews business information and supporting details.
    - Admin approves or rejects the request with a specified reason.
    - Business owner is notified of the decision.
    - Applications pending for more than **12 hours** trigger **automatic approval**, while admins are notified for record-keeping.

Business Rules:

- Rejected applications must include a rejection reason.
- Approved accounts gain full dashboard access immediately.
- Automatically approved accounts are flagged as “auto-approved” for traceability.

Acceptance Criteria:

- Admin can view all pending applications in a dedicated approval queue.
- Approval or rejection triggers automatic email notification.
- Approved users (manual or automatic) can log in immediately.

FR-BO-04: Onboarding Tutorial

- Priority: High
- Description: First-time business owners see a welcome screen with guided tour of key features.
- Features:
    - Welcome message
    - Quick tour highlighting: Add Product, View Orders, Access Analytics
    - Option to skip tour

        - "Get Started" checklist (Add first product, Set up payment info, Complete profile)
    - Acceptance Criteria:
        - Tour appears only on first login after approval
        - User can dismiss or skip tour
        - Checklist persists in dashboard until items completed

5.1.2 Dashboard (Overview)

FR-BO-05: Dashboard Home

- Priority: Critical
- Description: Business owner dashboard provides an at-a-glance view of business performance and quick access to key functions.
- Components:
    - Welcome message with business owner name
    - Summary Cards:
        - Total Products
        - Pending Orders
        - Revenue (Current Month)
        - Total Reviews/Average Rating
    - Quick Action Buttons:
        - Add New Product
        - View Raw Material Store
        - Check Recent Orders
        - View Financial Insights
    - Recent Activity Feed (last 10 actions)
    - Important Alerts Section (low stock, reviews, etc.)
- Acceptance Criteria:
    - Dashboard loads within 2 seconds
    - All metrics update in real-time or near real-time
    - Quick action buttons navigate to correct pages

FR-BO-06: Performance Metrics

- Priority: High
- Description: Display key performance indicators with visual charts.
- Metrics:
    - Sales trend (last 30 days) - line chart
    - Top 5 selling products - bar chart
    - Order status breakdown - pie chart
    - Revenue vs. expenses - comparison chart
- Acceptance Criteria:
    - Charts are interactive (hover for details)
    - Data refreshes when page is reloaded
    - Export option available for reports

5.1.3 Product Management

FR-BO-07: Product Listing

- Priority: Critical
- Description: Business owners can view all their products in a searchable, filterable table.
- Display Fields:
    - Product image (thumbnail)
    - Product name
    - Price
    - Stock quantity
    - Status (Draft / Pending Approval / Approved / Rejected)
    - Actions (Edit, Delete, View)
- Features:
    - Search by product name
    - Filter by status
    - Sort by date added, price, stock
    - Pagination (20 items per page)
- Acceptance Criteria:
    - Table displays all products owned by logged-in user
    - Search and filters work correctly
    - Status badges use distinct colors

FR-BO-08: Add New Product

- Priority: Critical
- Description: Business owners can create new product listings.
- Form Fields:
    - Product name (required, max 100 characters)
    - Category (dropdown from business type)
    - Description (required, rich text editor, max 2000 characters)
    - Price (required, decimal, > 0)
    - Stock quantity (required, integer, ≥ 0)
    - Product images (1-5 images, max 5MB each, JPEG/PNG)
    - Tags (optional, comma-separated) for product search and filtering
    - Variants (optional: size, color, etc.)
- Processing:
    - Validate all inputs
    - Upload images to storage
    - Save product with status "Pending Approval"
    - Notify admin of new submission
- Acceptance Criteria:
    - Form validates in real-time
    - Images preview before upload
    - Success message confirms submission
    - Product appears in seller's list with "Pending" status

FR-BO-09: Edit Product

- Priority: Critical
- Description: Business owners can modify existing product details.

- Rules:
    - Only products with status "Draft" or "Approved" can be edited
    - Editing an "Approved" product changes status back to "Pending Approval"
    - Cannot edit products with active orders (stock quantity can be updated)
- Acceptance Criteria:
    - Edit form pre-populates with current data
    - Changes save successfully
    - Status updates appropriately

FR-BO-10: Delete Product

- Priority: High
- Description: Business owners can delete products from their catalog.
- Rules:
    - Cannot delete products with pending or active orders
    - Deletion is soft delete (marked as deleted but retained in database)
    - Confirmation modal required before deletion
- Acceptance Criteria:
    - Confirmation asks "Are you sure?"
    - Products with active orders show error message
    - Deleted products no longer appear in seller's list

FR-BO-11: Product Analytics

- Priority: Medium
- Description: View performance metrics for each product.
- Metrics:
    - Total views
    - Add to cart count
    - Purchase count
    - Conversion rate
    - Revenue generated
- Acceptance Criteria:
    - Metrics accessible from product detail view
    - Data displays in visual format (charts/graphs)

5.1.4 Order Management

FR-BO-12: Order List

- Priority: Critical
- Description: Display all orders received by the business owner.
- Display Fields:
    - Order ID
    - Customer name
    - Order date
    - Total amount
    - Status (Pending / Processing / Shipped / Delivered / Refunded)
    - Actions (View Details, Update Status)
- Features:

        - Filter by status
        - Search by order ID or customer name
        - Sort by date, amount
        - Pagination
    - Acceptance Criteria:
        - Shows only orders for seller's products
        - Filters and search work correctly
        - Real-time status updates

FR-BO-13: Order Details

- Priority: Critical
- Description: View complete information about a specific order.
- Information Displayed:
    - Order ID and date
    - Customer information (name, phone,email)
    - Delivery address
    - Items ordered (product name, quantity, price)
    - Subtotal, shipping, total amount
    - Payment method and status
    - Order status and history
- Actions Available:
    - Update order status
    - Print invoice/packing slip
    - Contact customer (email/phone)
    - Issue refund (if applicable)
- Acceptance Criteria:
    - All order information displays correctly
    - Actions available based on current order status
    - Status history shows timestamps

FR-BO-14: Update Order Status

- Priority: Critical
- Description: Business owners can update the status of orders as they progress through fulfillment.
- Status Workflow:
    - Pending → Processing (when seller confirms order)
    - Processing → Shipped (when order is dispatched)
    - Shipped → Delivered (when customer receives order)
    - Any status → Refunded (in case of returns/cancellations)
- Processing:
    - Status update triggers notification to customer
    - Status change is logged with timestamp
    - Certain status changes may require additional info (tracking number for "Shipped")
- Acceptance Criteria:
    - Only valid status transitions are allowed

        - Customer receives notification within 1 minute
        - Optional tracking number field for "Shipped" status

FR-BO-15: Refund Processing

- Priority: High
- Description: Business owners can initiate refunds for orders.
- Process:
    - Select order for refund
    - Enter refund reason
    - Choose full or partial refund
    - Confirm refund
    - System processes refund through payment gateway
- Rules:
    - Only orders with status "Delivered" or "Shipped" can be refunded
    - Refund reason is mandatory
    - Customer notified automatically
- Acceptance Criteria:
    - Refund processes within 5-7 business days
    - Order status updates to "Refunded"
    - Financial records updated

5.1.5 Raw Material Store

FR-BO-16: Browse Raw Materials

- Priority: Critical
- Description: Business owners can browse available raw materials relevant to their business category.
- Display:
    - Grid layout with material cards “View in page”
    - Each card shows: image, name, supplier, unit price, unit of measure
- Features:
    - Filter by material category
    - Filter by supplier
    - Search by material name
    - Sort by price, supplier name(alphabetical)
- Acceptance Criteria:
    - Materials displayed match seller's business category
    - Grid is responsive across devices “Desketop , Mobile .....”
    - Images load efficiently

FR-BO-17: Material Details

- Priority: High
- Description: View detailed information about a specific raw material.
- Information Displayed:
    - Material name and description

        - Supplier name and (contact information is hidden until an order is added or hidden all time for BO)
        - Unit price and minimum order quantity
        - Available stock (hidden for BO)
        - Product images
        - Specifications (dimensions, weight, etc.)
    - Actions:
        - Add to cart
        - View supplier profile
        - Contact supplier (future enhancement)
    - Acceptance Criteria:
        - All material information displays correctly
        - Add to cart updates cart count immediately

FR-BO-18: Material Cart & Checkout

- Priority: Critical
- Description: Business owners can add materials to cart and complete purchase.
- Cart Features:
    - View all items in cart
    - Update quantities
    - Remove items
    - See running total
- Checkout Process:
    - Review order summary
    - Enter/confirm delivery address
    - Select payment method
    - Confirm and pay
- Processing:
    - Create order record
    - Process payment via Stripe
    - Upon successful checkout, a confirmation notification shall be sent to the seller. In the MVP, supplier notifications will be handled manually by the Admin, who will contact suppliers directly.
    - Record expense in financial system
- Acceptance Criteria:
    - Cart persists across sessions
    - Payment processes securely
    - Order confirmation sent immediately
    - Material expenses appear in Financial Insights

FR-BO-19: Material Order History

- Priority: Medium
- Description: View history of all raw material purchases.
- Display:
    - Order ID, date, supplier, total amount, status
    - Filter by date range, supplier, status

- Acceptance Criteria:
    - All material orders are logged
    - Searchable and filterable
    - Can reorder from history

5.1.6 Financial Insights

FR-BO-20: Financial Dashboard

- Priority: Critical
- Description: Provide comprehensive view of business financial performance.
- Components:
    - Summary Cards:
        - Total Revenue (customizable period)
        - Total Expenses
        - Net Profit
        - Pending Payouts
    - Revenue Chart (line graph over time)
    - Expense Breakdown (pie chart by category)
    - Profit Trend (bar chart monthly comparison)
- Features:
    - Date range selector (last 7 days, 30 days, 3 months, custom range)
    - Export to PDF/CSV
    - Print report
- Acceptance Criteria:
    - All figures accurate and up-to-date
    - Charts are interactive
    - Exports include all selected data

FR-BO-21: Transaction History

- Priority: High
- Description: Detailed list of all financial transactions.
- Display Fields:
    - Date
    - Type (Sale / Material Purchase / Refund / Payout / Fee)
    - Description
    - Amount
    - Balance after transaction
- Features:
    - Filter by type, date range
    - Search by description
    - Pagination
- Acceptance Criteria:
    - All transactions logged accurately
    - Running balance calculated correctly
    - Filters work as expected

FR-BO-22: Payout Management

- Priority: High
- Description: Business owners can view available balance and request payouts.
- Features:
    - Display available balance (revenue minus fees/expenses)
    - Display pending payout requests
    - Request payout button
    - View payout history
- Payout Process:
    - Enter payout amount (must be ≤ available balance)
    - Confirm bank details
    - Submit request
    - Admin reviews and processes
    - Notification sent when completed
- Business Rules:
    - Minimum payout amount: [SPECIFY AMOUNT]
    - Payout processing time: 3-5 business days
    - Maximum one pending request at a time
- Acceptance Criteria:
    - Available balance is accurate
    - Cannot request more than available
    - Request confirmation sent immediately

FR-BO-23: Export Financial Reports

- Priority: Medium
- Description: Generate and download financial reports in various formats.
- Report Types:
    - Sales summary
    - Expense report
    - Profit & loss statement
    - Tax report (basic)
- Formats:
    - PDF (formatted for printing)
    - CSV (for spreadsheet import)
    - Excel (formatted with charts)
- Acceptance Criteria:
    - Reports generate within 10 seconds
    - Data matches dashboard figures
    - Files download successfully

5.1.7 Ratings & Reviews

FR-BO-24: View Reviews

- Priority: High
- Description: Business owners can view all customer reviews for their products.
- Display:
    - Review cards showing:

        - Customer name (or anonymous if requested)
        - Rating (stars)
        - Review text
        - Product name
        - Date posted
        - Seller response (if any)
- Features:
    - Filter by rating (1-5 stars)
    - Filter by product
    - Sort by date, rating
    - Search review text
- Acceptance Criteria:
    - All reviews display correctly
    - Filters work accurately
    - Unresponded reviews highlighted

FR-BO-25: Respond to Reviews

- Priority: High
- Description: Business owners can reply to customer reviews.
- Process:
    - Click "Respond" on review
    - Enter response text (max 500 characters)
    - Submit response
    - Response appears under original review
- Rules:
    - One response per review
    - Response can be edited within 24 hours of posting
    - Response is public
- Acceptance Criteria:
    - Response saves and displays immediately
    - Customer receives notification of response
    - Edit option available for 24 hours

FR-BO-26: Review Analytics

- Priority: Medium
- Description: View aggregate review statistics.
- Metrics:
    - Overall average rating
    - Rating distribution (5-star, 4-star, etc.)
    - Total number of reviews
    - Response rate
    - Sentiment trend over time
- Acceptance Criteria:
    - Metrics update automatically with new reviews
    - Visual charts display distribution

5.1.8 Support & Services

FR-BO-27: Support Ticket System

- Priority: High
- Description: Business owners can submit and track support tickets.
- Ticket Creation:
    - Select category (Technical / Account / Payment / Other)
    - Enter subject line
    - Describe issue (rich text editor)
    - Attach files (optional, max 3 files, 10MB each)
    - Submit ticket
- Ticket Management:
    - View all submitted tickets
    - Check ticket status (Open / In Progress / Resolved / Closed)
    - Add follow-up messages
    - Close ticket when resolved
- Acceptance Criteria:
    - Ticket submission confirms with ticket number
    - Email notification sent on status changes
    - Can attach screenshots/documents

FR-BO-28: Request Growth Services

- Priority: Medium (MVP), High (Phase 2)
- Description: Business owners can request additional services like branding, analytics, or marketing support.
- Service Types:
    - Branding consultation
    - Advanced analytics
    - Marketing campaign support
    - Photography services
- Process:
    - Browse available services
    - Select service
    - Fill service request form
    - Submit request
    - Admin reviews and contacts seller
- Acceptance Criteria:
    - Services display with descriptions and pricing
    - Request form captures necessary information
    - Confirmation sent upon submission

FR-BO-29: FAQ Access

- Priority: Medium
- Description: Quick access to frequently asked questions.
- Features:
    - Searchable FAQ database
    - Categories (Getting Started / Products / Orders / Payments)

        - Related articles suggestions
    - Acceptance Criteria:
        - FAQs are well-organized and easy to find
        - Search returns relevant results
        - Links to create ticket if FAQ doesn't help

5.1.9 Notifications

FR-BO-30: Notification Center

- Priority: High
- Description: Centralized location for all notifications.
- Notification Types:
    - System notifications (maintenance, updates)
    - Order notifications (new order, status changes)
    - Financial notifications (payout processed, low balance)
    - Support notifications (ticket responses)
    - Product notifications (approval, rejection, low stock)
- Features:
    - Badge showing unread count
    - Mark as read/unread
    - Clear all read notifications
    - Filter by category
- Delivery Channels:
    - In-app (notification center)
    - Email (for critical notifications)
    - Real-time via SignalR
- Acceptance Criteria:
    - Notifications appear in real-time
    - Unread count is accurate
    - Clicking notification navigates to relevant page

FR-BO-31: Notification Preferences

- Priority: Medium
- Description: Business owners can customize notification settings.
- Options:
    - Enable/disable email notifications by category
    - Set quiet hours for non-critical notifications
    - Choose notification sound
- Acceptance Criteria:
    - Preferences save successfully
    - Notifications respect user preferences
    - Critical notifications always delivered

5.1.10 Account Settings

FR-BO-32: Profile Management

- Priority: High
- Description: Business owners can update personal and business information.
- Editable Fields:
    - Personal: Name, email, phone, profile photo
    - Business: Business name, description, logo, address
    - Contact: Social media links, website URL
- Rules:
    - Email change requires verification
    - Business name must remain unique
    - Profile photo max 2MB
- Acceptance Criteria:
    - Changes save successfully
    - Confirmation message displays
    - Email verification sent if email changed

FR-BO-33: Payment Information

- Priority: Critical
- Description: Manage bank account details for payouts.
- Information:
    - Bank name
    - Account holder name
    - Account number
    - Routing/SWIFT code
- Security:
    - Account number partially masked when displayed
    - Password re-entry required to view/edit
    - Changes logged for security
- Acceptance Criteria:
    - Information encrypted in database
    - Masked display for security
    - Validation prevents invalid account numbers

FR-BO-34: Security Settings

- Priority: High
- Description: Manage account security options.
- Features:
    - Change password
    - Enable/disable two-factor authentication (2FA)
    - View login history
    - Active sessions management (logout other devices)
- 2FA Implementation:
    - SMS-based or authenticator app
    - Required for sensitive operations when enabled
- Acceptance Criteria:
    - Password change requires current password
    - 2FA setup is straightforward

        - Login history shows device and location

FR-BO-35: Preferences

- Priority: Low
- Description: Customize platform experience.
- Options:
    - Language (for Phase 2)
    - Timezone
    - Date format
    - Currency display
    - Dashboard layout preferences
- Acceptance Criteria:
    - Preferences persist across sessions
    - Changes apply immediately

5.1.11 Knowledge Base

FR-BO-36: Browse Educational Content

- Priority: Medium
- Description: Access guides, tutorials, and educational materials.
- Content Types:
    - Getting started guides
    - Video tutorials
    - Best practices articles
    - Business tips
    - Platform updates
- Features:
    - Search functionality
    - Category browsing
    - Tag filters
    - Bookmark favorites
- Acceptance Criteria:
    - Content is well-organized and searchable
    - Videos open in a new tab or redirect to external source
    - PDFs open in new tab or download

FR-BO-37: Content Recommendations

- Priority: Low
- Description: Suggest relevant educational content based on user behavior.
- Logic:
    - New users see "Getting Started" content
    - Users with low sales see "Marketing Tips"
    - Users with many products see "Inventory Management"
- Acceptance Criteria:
    - Recommendations appear on dashboard
    - Suggestions are contextually relevant

5.2 Customer Module

5.2.1 Home Page

FR-CU-01: Homepage Layout

- Priority: Critical
- Description: The landing page provides an engaging entry point to the marketplace.
- Components:
    - Hero banner with call-to-action
    - Featured brands carousel
    - Trending products grid
    - Category cards (Fashion, Crafts, Beauty)
    - Search bar (prominent placement)
    - Testimonials/reviews section
    - Newsletter signup
- Acceptance Criteria:
    - Page loads within 3 seconds
    - All sections are responsive
    - CTA buttons navigate correctly

FR-CU-02: Product Search

- Priority: Critical
- Description: Customers can search for products across the entire marketplace.
- Search Features:
    - Text input with autocomplete suggestions
    - Search by product name, description, tags
    - Search results page with filters
- Filters:
    - Category
    - Price range (slider)
    - Rating (minimum stars)
    - Brand
    - Availability (in stock only)
- Sorting Options:
    - Relevance (default)
    - Price: Low to High
    - Price: High to Low
    - Newest First
    - Best Rated
- Acceptance Criteria:
    - Search returns results within 1 second
    - Autocomplete suggests relevant terms
    - Filters refine results accurately

FR-CU-03: Category Browsing

- Priority: High

- Description: Browse products by business category.
- Categories:
    - Fashion & Accessories
    - Handmade Crafts
    - Natural & Beauty Products
- Display:
    - Category page shows all products in that category
    - Subcategory filters (if applicable)
    - Same filtering and sorting as search
- Acceptance Criteria:
    - Category pages load quickly
    - Product counts are accurate
    - Filters work correctly

FR-CU-04: Featured & Trending Products

- Priority: Medium
- Description: Highlight selected products on the homepage.
- Featured Products:
    - Admin-selected products for promotion
    - Displayed in prominent carousel
- Trending Products:
    - Based on recent sales, views, and engagement
    - Updates automatically based on metrics
- Acceptance Criteria:
    - Featured products rotate every 5 seconds
    - Trending section updates daily
    - Clicking navigates to product details

5.2.2 Brand Exploration

FR-CU-05: Brand Listing

- Priority: High
- Description: View all business owners (brands) on the platform.
- Display:
    - Grid of brand cards showing:
        - Brand logo
        - Brand name
        - Category
        - Average rating
        - Number of products
- Features:
    - Search by brand name
    - Filter by category
    - Sort by rating, product count, newest
- Acceptance Criteria:
    - Only approved brands appear

        - Brand cards link to brand pages
        - Filtering works correctly

FR-CU-06: Brand Page

- Priority: Critical
- Description: Dedicated page showcasing a specific brand and their products.
- Components:
    - Brand header:
        - Logo and banner image
        - Brand name and description
        - Social media links
        - Average rating and review count
    - Product grid (all products from this brand)
    - About section
    - Customer reviews
- Features:
    - Filter products by subcategory
    - Sort products by price, popularity
    - Follow brand (wishlist all products - future)
- Acceptance Criteria:
    - Page displays brand info accurately
    - All active products shown
    - Social links open in new tab

5.2.3 Product Discovery

FR-CU-07: Product Details Page

- Priority: Critical
- Description: Comprehensive view of a single product with all relevant information.
- Components:
    - Image gallery (with zoom and thumbnail navigation)
    - Product title
    - Brand name (linked to brand page)
    - Price
    - Stock availability indicator
    - Product description (formatted text)
    - Specifications/details
    - Ratings summary and reviews
    - Similar products section
    - Action buttons:
        - Add to Cart
        - Buy Now
        - Add to Wishlist
- Variant Selection:
    - If product has variants (size, color), show selection dropdowns
    - Price and stock update based on variant selection

- Acceptance Criteria:
    - All product information displays correctly
    - Images load quickly and support zoom
    - Variant selection updates price/stock
    - Out of stock products cannot be added to cart

FR-CU-08: Product Reviews Section

- Priority: High
- Description: Display customer reviews and ratings for a product.
- Display:
    - Overall rating (stars and numeric)
    - Rating distribution (5-star breakdown)
    - Review cards:
        - Customer name
        - Rating
        - Review text
        - Date posted
        - Seller response (if any)
        - Helpful votes
- Features:
    - Sort reviews (Most Recent / Highest Rated / Lowest Rated)
    - Filter by star rating
    - Pagination (10 reviews per page)
    - Mark review as helpful
- Acceptance Criteria:
    - Reviews display in readable format
    - Average rating matches individual ratings
    - Sorting and filtering work correctly

FR-CU-09: Write Product Review

- Priority: High
- Description: Customers can write reviews for products they have purchased.
- Requirements:
    - Must be logged in
    - Must have purchased the product
    - Can only review once per product
- Review Form:
    - Star rating (1-5, required)
    - Review title (optional, max 100 characters)
    - Review text (required, max 1000 characters)
    - Upload photos (optional, max 3 images)
- Processing:
    - Validate customer eligibility
    - Save review
    - Update product average rating
    - Notify business owner

- Acceptance Criteria:
    - Only eligible customers can access review form
    - Review appears immediately after submission
    - Business owner receives notification

5.2.4 Cart & Checkout

FR-CU-10: Shopping Cart

- Priority: Critical
- Description: Customers shall be able to add products to a shopping cart and manage items before proceeding to checkout. Only logged-in users can add products to the cart.
- Cart Features:
    - Add products with quantity selection
    - View all items in cart
    - Update item quantities
    - Remove items
    - View subtotal, estimated shipping, and total
- Cart Rules:
    - Cart persists for logged-in users across sessions
    - Stock availability checked in real-time
- Display:
    - Cart icon with item count badge in header
    - Cart page with full item details:
        - Product image and name
        - Variant details (if applicable)
        - Unit price
        - Quantity selector
        - Line total
        - Remove button
- Acceptance Criteria:
    - Cart updates immediately when items added/removed
    - Quantity changes update total in real-time
    - Out of stock items show warning
    - Only logged-in users can add products to the cart

FR-CU-11: Proceed to Checkout

- Priority: Critical
- Description: Navigate from cart to checkout process.
- Pre-checkout Validations:
    - All items in stock
    - Cart not empty
- Acceptance Criteria:
    - Button disabled if validations fail
    - Clear error messages if issues exist
    - Redirects to checkout page when valid

FR-CU-12: Checkout Process

- Priority: Critical
- Description: Multi-step checkout flow to complete purchase.
- Step 1: Delivery Information
    - Full name
    - Phone number
    - Delivery address (street, city, postal code, country)
    - Save address for future use (checkbox)
- Step 2: Review Order
    - Order summary (items, quantities, prices)
    - Delivery address confirmation
    - Estimated delivery date
    - Shipping cost
    - Total amount
- Step 3: Payment
    - Payment method selection:
        - Credit/Debit Card (via Stripe)
        - Cash on Delivery (if enabled)
    - Stripe payment form (if card selected)
- Step 4: Confirmation
    - Order ID
    - Estimated delivery date
    - Payment confirmation
    - Order tracking link
    - "Continue Shopping" button
- Processing:
    - Create order records in database
    - Process payment via Stripe
    - Send confirmation emails (customer and seller)
    - Trigger notification to seller
    - Empty cart
- Acceptance Criteria:
    - Checkout flow is intuitive and smooth
    - Payment processes securely through Stripe
    - Confirmation email sent within 1 minute
    - Order appears in customer's order history

FR-CU-13: Guest Checkout
Priority: Medium
Description: Non-registered users (guests) shall be able to browse and view products, but they **cannot add products to the cart or complete purchases**.

Features:

- Browse and view product details
- Search and filter products

- View product images, descriptions, prices, and specifications

Restrictions:

- Guests cannot add products to the cart
- Guests cannot proceed to checkout or make a purchase
- Call-to-action for purchasing prompts guest to **register or log in**

Acceptance Criteria:

- Guests can view all available products
- Attempting to add a product to the cart redirects or prompts for login/registration
- All product information displays correctly for guests

5.2.5 Wishlist

FR-CU-14: Add to Wishlist

- Priority: High
- Description: Customers can save products to a wishlist for future reference.
- Functionality:
    - Heart icon on product cards and detail pages
    - Click to add/remove from wishlist
    - Visual indication when product is wishlisted
- Requirements:
    - Must be logged in to use wishlist
    - No limit on wishlist items
- Acceptance Criteria:
    - Icon updates immediately when clicked
    - Wishlist persists across sessions

FR-CU-15: View Wishlist

- Priority: High
- Description: Dedicated page showing all wishlisted products.
- Display:
    - Grid layout similar to product browsing
    - Each item shows: image, name, price, stock status
    - Actions: Add to Cart, Remove from Wishlist
- Features:
    - Move multiple items to cart at once
    - Remove items individually
    - See price changes (if product price updated)
    - Stock status indicators
- Acceptance Criteria:
    - All wishlisted products display
    - Add to cart works directly from wishlist
    - Out of stock items clearly marked

5.2.6 Order Tracking

FR-CU-16: Order History

- Priority: Critical
- Description: Customers can view all their past and current orders.
- Display:
    - List/grid of orders showing:
        - Order ID
        - Order date
        - Total amount
        - Status
        - Actions (View Details, Track, Review Products)
- Features:
    - Filter by status (All / Pending / Shipped / Delivered)
    - Filter by date range
    - Search by order ID
- Acceptance Criteria:
    - All orders for logged-in customer display
    - Orders sorted by date (newest first)
    - Filters work correctly

FR-CU-17: Order Details

- Priority: Critical
- Description: View complete information about a specific order.
- Information Displayed:
    - Order ID and date
    - Items ordered (product name, image, quantity, price)
    - Delivery address
    - Payment method
    - Order timeline/status history
    - Tracking information (if shipped)
    - Total amount breakdown
- Actions Available:
    - Track shipment
    - Request return/refund (within return window)
    - Download invoice (if Delivered after 2 or 3 days)
- Acceptance Criteria:
    - All order details accurate
    - Status timeline shows progression
    - Actions available based on order status

FR-CU-18: Shipment Tracking

- Priority: High
- Description: Track the delivery status of shipped orders.
- Display:
    - Tracking number
    - Current status

        - Status timeline (Order Confirmed → Shipped → Out for Delivery → Delivered)
        - Estimated delivery date
        - Actual delivery date (once delivered)
    - Acceptance Criteria:
        - Tracking updates reflect current status
        - Timeline visualization is clear
        - Delivery date estimations are reasonable

FR-CU-19: Order Cancellation

- Priority: High
- Description: Customers can cancel orders before they ship.
- Rules:
    - Only orders with status "Pending" can be cancelled
    - Cancellation triggers automatic refund
    - Seller notified of cancellation
- Process:
    - Click "Cancel Order"
    - Confirm cancellation
    - Order status updates to "Cancelled"
    - Refund initiated
- Acceptance Criteria:
    - Cannot cancel shipped orders
    - Confirmation modal prevents accidental cancellations
    - Refund processed within specified timeframe (ex: 3 or 4 days)

5.2.7 Customer Notifications

FR-CU-20: Order Notifications

- Priority: High
- Description: Customers receive notifications about order status changes.
- Notification Triggers:
    - Order confirmed
    - Payment received
    - Order shipped (includes tracking info)
    - Out for delivery
    - Order delivered
    - Refund processed
- Delivery Channels:
    - In-app notification center
    - Email
    - Real-time via SignalR
- Acceptance Criteria:
    - Notifications sent within 1 minute of status change
    - Email includes relevant order details
    - In-app notifications are actionable (click to view order)

FR-CU-21: Promotional Notifications

- Priority: Low
- Description: Customers receive notifications about promotions and new products.
- Types:
    - New arrivals from followed brands (future)
    - Flash sales and promotions
    - Back-in-stock alerts for wishlisted items
- User Control:
    - Opt-in/opt-out in account settings
    - Frequency preferences
- Acceptance Criteria:
    - Respects user preferences
    - Unsubscribe link in promotional emails
    - Not intrusive to user experience

5.3 Admin Module

5.3.1 Dashboard & Analytics

FR-AD-01: Admin Dashboard

- Priority: Critical
- Description: High-level overview of platform health and performance.
- Key Metrics Cards:
    - Total Sellers (Active / Pending / Suspended)
    - Total Customers
    - Total Products (Approved / Pending)
    - Total Orders (by status)
    - Platform Revenue (current month)
    - Pending Actions Count
- Charts:
    - Revenue trend (last 6 months)
    - User growth (sellers and customers over time)
    - Order volume trend
    - Category distribution (products by category)
- Important Alerts Section:
    - Products with low stock
    - Sellers awaiting approval before automatic approval
    - Unresolved complaints > 48 hours
    - System errors or issues
- Recent Activity Feed:
    - New seller registrations

        - New products submitted
        - Recent orders
        - Support tickets created
- Acceptance Criteria:
        - Dashboard loads within 3 seconds
        - All metrics are accurate and real-time
        - Alerts are actionable (click to navigate)

FR-AD-02: Platform KPIs

- Priority: High
- Description: Detailed view of key performance indicators.
- Metrics:
        - Conversion rate (visitors to buyers)
        - Average order value
        - Customer retention rate
        - Seller churn rate
        - Product approval rate
        - Average time to ship
        - Customer satisfaction score
- Visualization:
        - Line charts for trends
        - Gauge charts for percentages
        - Comparison to previous period
- Acceptance Criteria:
        - KPIs calculate correctly
        - Historical data available
        - Export functionality for reports (ex : Excel , CSV , PDF ....)

5.3.2 User Management

FR-AD-03: Seller Management

- Priority: Critical
- Description: View and manage all business owner accounts.
- Display:
        - Table with columns:
                - Seller ID
                - Business Name
                - Owner Name
                - Email
                - Category
                - Status (Pending / Active / Suspended / Banned)
                - Registration Date
                - Actions
- Features:
        - Search by name, email, business name
        - Filter by status, category, registration date

        - Sort by any column
        - Pagination
    - Actions:
        - View seller details
        - Approve pending seller
        - Reject seller application
        - Suspend account
        - Ban account
        - View seller's products
        - View seller's orders
        - Contact seller
    - Acceptance Criteria:
        - All sellers displayed correctly
        - Filters and search work accurately
        - Actions execute successfully with confirmation

FR-AD-04: Approve/Reject Sellers

- Priority: Critical
- Description: Review and approve or reject seller applications.
- Approval Process:
    - View seller's business information
    - Verify business details
    - Check for policy compliance
    - Approve or reject with reason
    - Notification sent to seller
- Approval Criteria:
    - Complete business information
    - Valid contact details
    - Appropriate business category
    - No duplicate business names
- Rejection Reasons:
    - Incomplete information
    - Inappropriate business type
    - Duplicate account
    - Policy violation
- Acceptance Criteria:
    - Admin can review all pending applications
    - Approval grants full dashboard access
    - Rejection includes clear reason
    - Email notification sent immediately

FR-AD-05: Suspend/Ban Sellers

- Priority: High
- Description: Take disciplinary action against sellers violating policies.
- Suspension:
    - Temporary account restriction

        - Seller cannot log in
        - Products hidden from marketplace
        - Existing orders remain active for fulfillment
        - Can be reversed by admin
    - Ban:
        - Permanent account termination
        - All products permanently hidden
        - Cannot register with same email/business name
        - Cannot be reversed without Super Admin approval
    - Process:
        - Select seller account
        - Choose action (Suspend / Ban)
        - Enter reason (mandatory)
        - Confirm action
        - System logs action with timestamp and admin ID
        - Notification sent to seller
    - Acceptance Criteria:
        - Actions execute immediately
        - Seller cannot access account when suspended/banned
        - All actions logged for audit trail
        - Clear notification sent with explanation

FR-AD-06: Buyer Management

- Priority: High
- Description: View and manage customer accounts.
- Display:
    - Table with columns:
        - Customer ID
        - Name
        - Email
        - Registration Date
        - Total Orders
        - Total Spent
        - Status
        - Actions
- Features:
    - Search by name, email
    - Filter by status, registration date, order count
    - Sort by any column
- Actions:
    - View customer details
    - View order history
    - View reviews written

        - Block account with reason written .
        - Delete account (with confirmation)
        - View as customer (simulate customer view)
    - Acceptance Criteria:
        - All customers displayed
        - Order history and stats accurate
        - Actions require confirmation for safety

FR-AD-07: Customer Experience Preview

- Priority: Medium
- Description: Admin can view a read-only version of a buyer's marketplace experience.
- Purpose:
    - Troubleshoot user-reported issues
    - Understand customer perspective
    - Verify display of products/brands
- Implementation:
    - Button in buyer details: "View as Customer"
    - Opens customer's view in new tab
    - Read-only mode (cannot place orders)
    - Banner indicates admin preview mode
- Acceptance Criteria:
    - Displays customer's actual view
    - Cannot perform transactions
    - Easy to exit preview mode

FR-AD-08: Policy Violation Handling

- Priority: High
- Description: Automated and manual handling of policy violations.
- Auto-Block Rules:
    - Seller receives 3 confirmed complaints → Auto-block + Admin notification
    - Account inactive > 12 months → Auto-deactivate
    - Detection of fake/duplicate accounts → Flag for admin review
- Manual Review Process:
    - Admin receives notification of auto-block
    - Reviews user activity and violation details
    - Decides action: Maintain Block / Warn / Investigate / Unblock
    - Records decision with notes
- Complaint Management:
    - Complaint ticket created by buyer/seller
    - Linked to relevant user accounts
    - Status progression: Open → In Progress → Resolved
    - Resolution noted with action taken
- Acceptance Criteria:
    - Auto-block triggers work correctly
    - Admin notified immediately
    - All decisions logged with timestamp and reason

        - Users notified of actions taken

5.3.3 Product & Service Management

FR-AD-09: Product Approval Queue

- Priority: Critical
- Description: Review and approve/reject product submissions from sellers.
- Display:
    - List of pending products with:
        - Product image
        - Product name
        - Seller name
        - Category
        - Price
        - Submission date
        - Quick actions
- Review Process:
    - Click product to view full details
    - Review product information:
        - Images (check quality and appropriateness)
        - Description (check for completeness and policy compliance)
        - Pricing (check for reasonableness)
        - Category assignment
    - Make decision: Approve / Reject / Request Changes
    - If rejecting, provide clear reason
- Rejection Reasons:
    - Poor image quality
    - Incomplete description
    - Inappropriate content
    - Wrong category
    - Price violation
    - Duplicate listing
- Bulk Actions:
    - Select multiple products
    - Approve all selected
    - Reject all selected (with common reason)
- Acceptance Criteria:
    - All pending products visible to admin
    - Approval makes product live immediately
    - Rejection sends notification to seller with reason
    - Bulk actions work correctly

FR-AD-10: Product Moderation

- Priority: High
- Description: Monitor and manage all products on the platform.
- View All Products:

        - Table showing all products (approved, pending, rejected)
        - Filters: Status, Category, Seller, Price Range, Date
        - Search by product name
    - Actions:
        - View product details
        - Approve or reject product edits requested by sellers
        - Change category (if misclassified)
        - Temporarily hide or remove products violating platform policies (soft delete)
        - Feature product on homepage
        - View product analytics
    - Analytics per Product:
        - Total views
        - Add to cart count
        - Purchase count
        - Conversion rate
        - Revenue generated
    - Acceptance Criteria:
        - Admin can view all products across platform
        - Editing preserves data integrity
        - Removed products hidden from marketplace
        - Analytics accurate

FR-AD-11: Low Stock Alerts

- Priority: Medium
- Description: Monitor products with low inventory levels.
- Alert Criteria:
    - Stock falls below predefined threshold (default: 5 units)
    - Threshold configurable per category
- Alert Display:
    - Dashboard alert section shows count
    - Dedicated page lists all low-stock products
    - Includes: Product name, Seller, Current stock, Last sale date
- Actions:
    - Notify seller (manual or automatic)
    - View product details
    - Contact seller
- Acceptance Criteria:
    - Alerts trigger at correct threshold
    - List updates in real-time
    - Notifications sent to sellers

FR-AD-12: Service Request Management

- Priority: Medium
- Description: Handle seller requests for platform-provided services.
- Service Types:
    - Branding consultation

        - Advanced analytics access
        - Marketing campaign support
        - Professional photography
    - Process:
        - View all service requests
        - Review request details and seller profile
        - Assign to team member (if applicable)
        - Communicate with seller
        - Update request status
        - Mark as completed
    - Acceptance Criteria:
        - All requests visible and trackable
        - Status updates notify seller
        - Completed requests archived

5.3.4 Orders & Transactions

FR-AD-13: Order Overview

- Priority: Critical
- Description: Monitor all orders across the platform.
- Display:
    - Comprehensive table with:
        - Order ID
        - Order Date
        - Customer Name
        - Seller Name
        - Total Amount
        - Status
        - Payment Status
        - Actions
- Features:
    - Search by Order ID, customer, seller
    - Filter by status, payment status, date range
    - Sort by date, amount
    - Export filtered results
- Statistics:
    - Total orders (by status)
    - Total revenue
    - Average order value
    - Orders by category
- Acceptance Criteria:
    - All platform orders visible
    - Real-time updates
    - Filters and search accurate
    - Export includes all visible data

FR-AD-14: Order Details & Intervention

- Priority: High
- Description: View complete order information and intervene when necessary.
- Information Displayed:
    - Full order details (customer, seller, items, amounts)
    - Order timeline and status history
    - Payment information
    - Shipping/tracking details
    - Related tickets or complaints
- Admin Actions:
    - Contact customer or seller
    - Force status change (with reason)
    - Process refund
    - Cancel order
    - Add internal notes
- Use Cases:
    - Dispute resolution
    - Delayed shipments
    - Payment issues
    - Customer complaints
- Acceptance Criteria:
    - Complete order information accessible
    - Admin actions logged
    - Parties notified of admin intervention

FR-AD-15: Transaction Management

- Priority: High
- Description: Monitor payment transactions and resolve payment issues.
- Display:
    - Transaction log showing:
        - Transaction ID
        - Order ID
        - Amount
        - Payment Method
        - Status (Success / Failed / Pending / Refunded)
        - Date/Time
- Features:
    - Filter by status, payment method, date
    - Search by transaction or order ID
    - View transaction details
    - Retry failed payments (if applicable)
- Payment Disputes:
    - View disputed transactions
    - Review dispute details
    - Communicate with payment gateway

        - Resolve or escalate
    - Acceptance Criteria:
        - All transactions logged accurately
        - Failed transactions clearly marked
        - Dispute resolution tracked

FR-AD-16: Refund Processing

- Priority: High
- Description: Oversee and process refund requests.
- Refund Queue:
    - List of pending refund requests
    - Includes: Order ID, Customer, Amount, Reason, Request Date
- Review Process:
    - View order and refund details
    - Check refund eligibility
    - Approve or deny refund
    - If approved, process through payment gateway
    - Update order status
- Refund Rules:
    - Must be within return window (default: 14 days after delivery)
    - Product must be in acceptable condition
    - Seller consent obtained (or admin override)
- Acceptance Criteria:
    - Refund requests visible and trackable
    - Approval processes refund automatically
    - Customer and seller both notified

FR-AD-17: Financial Reports

- Priority: High
- Description: Generate comprehensive financial reports for the platform.
- Report Types:
    - Revenue Report (total revenue, by category, by seller)
    - Transaction Report (all transactions with details)
    - Payout Report (payouts to sellers)
    - Refund Report (all refunds processed)
    - Commission Report (platform fees collected)
- Features:
    - Date range selection
    - Export to CSV, Excel, PDF
    - Schedule automated reports (email delivery)
    - Visual charts and graphs
- Acceptance Criteria:
    - Reports generate accurate data
    - Export formats work correctly
    - Scheduled reports deliver on time

5.3.5 Reports & Analytics

FR-AD-18: Platform Analytics Dashboard

- Priority: High
- Description: Comprehensive analytics showing platform performance.
- Key Metrics:
    - User Metrics:
        - New registrations (sellers, customers)
        - Active users
        - User retention rate
        - Churn rate : is the percentage of users who stop using the platform over a specific period.
    - Sales Metrics:
        - Total sales volume
        - Sales by category
        - Average order value
        - Repeat purchase rate
    - Product Metrics:
        - Total products
        - Products by category
        - Top-performing products
        - Product approval rate
    - Engagement Metrics:
        - Average session duration
        - Pages per session
        - Bounce rate
        - Conversion rate
- Visualization:
    - Interactive charts and graphs
    - Time-based trends
    - Comparison views (current vs. previous period)
    - Drill-down capabilities
- Acceptance Criteria:
    - All metrics calculate accurately
    - Charts are interactive
    - Data refreshes regularly

FR-AD-19: Seller Performance Reports

- Priority: High
- Description: Analyze individual seller performance.
- Metrics per Seller:
    - Total products listed
    - Total sales volume
    - Total revenue
    - Average order fulfillment time
    - Customer rating

        - Return/refund rate
    - Ranking:
        - Top sellers by revenue
        - Top sellers by order count
        - Top rated sellers
        - Fastest growing sellers
    - Use Cases:
        - Identify star performers for featuring
        - Identify struggling sellers for support
        - Detect problematic sellers
    - Acceptance Criteria:
        - Seller rankings accurate
        - Can drill into individual seller details
        - Export seller reports

FR-AD-20: Customer Insights

- Priority: Medium
- Description: Understand customer behavior and preferences.
- Insights:
    - Customer lifetime value
    - Purchase frequency distribution
    - Average cart value
    - Most popular products
    - Peak shopping times
    - Geographic distribution
    - Category preferences
- Segmentation:
    - High-value customers
    - At-risk customers (no purchase in X days)
    - New vs. returning customers
- Acceptance Criteria:
    - Insights are actionable
    - Segmentation is accurate
    - Data respects privacy regulations

FR-AD-21: Category Performance

- Priority: Medium
- Description: Analyze performance by business category.
- Metrics per Category:
    - Number of sellers
    - Number of products
    - Total sales
    - Average price point
    - Customer ratings
    - Growth trend
- Comparison:

        - Side-by-side category comparison
        - Market share by category
    - Use Cases:
        - Identify which categories to expand
        - Allocate marketing resources
        - Plan inventory and suppliers
    - Acceptance Criteria:
        - All three categories tracked
        - Comparisons are visual and clear
        - Historical data available

5.3.6 Complaints & Support Center

FR-AD-22: Support Ticket Queue

- Priority: Critical
- Description: Centralized view of all support tickets from users.
- Display:
    - Ticket list showing:
        - Ticket ID
        - User Name and Type (Customer/Seller)
        - Category
        - Subject
        - Status (Open / In Progress / Resolved / Closed)
        - Priority (Low / Medium / High / Urgent)
        - Assigned To
        - Created Date
        - Last Updated
- Features:
    - Filter by status, category, priority, user type
    - Search by ticket ID, user name, keywords
    - Sort by date, priority
    - Assign tickets to admin staff
    - Bulk status updates
- Acceptance Criteria:
    - All tickets visible in queue
    - Filters work accurately
    - Assignment notifications sent

FR-AD-23: Ticket Management

- Priority: Critical
- Description: View and respond to individual support tickets.
- Ticket Details View:
    - Full ticket information
    - User profile summary
    - Complete message thread
    - Attached files/screenshots

        - Related orders or products (if applicable)
        - Internal notes (not visible to user)
    - Actions:
        - Reply to user
        - Update status
        - Change priority
        - Assign/reassign to another admin
        - Escalate to higher level
        - Add internal notes
        - Link to related tickets
        - Close/resolve ticket
    - Response Features:
        - Rich text editor
        - Attach files
        - Use templated responses
        - CC other staff members
    - Acceptance Criteria:
        - Complete ticket history visible
        - Responses send notifications to user
        - Internal notes remain private
        - Status updates logged

FR-AD-24: Complaint Resolution

- Priority: High
- Description: Handle complaints between buyers and sellers.
- Complaint Types:
    - Product quality issues
    - Shipping delays
    - Wrong item received
    - Seller unresponsive
    - Refund disputes
    - Policy violations
    - Resolution Process:
        - Review complaint details
        - View related order and product
        - Contact both parties
        - Gather evidence (photos, messages)
        - Make decision based on platform policies
        - Take action (refund, warning, block, etc.)
        - Document resolution
        - Close complaint
    - Escalation:
        - Complex cases escalate to senior admin
        - Legal issues flag for management
    - SLA:

        - Response within 24 hours
        - Resolution within 48 hours (or notify of delay)
    - Acceptance Criteria:
        - Complaints tracked separately from general tickets
        - All parties notified of resolution
        - Actions taken logged
        - Unresolved complaints > 48 hours trigger alert

FR-AD-25: Ticket Categories & Auto-Assignment

- Priority: Medium
- Description: Organize tickets and automatically route them.
- Categories:
    - Technical Issues
    - Account Problems
    - Payment Issues
    - Order Questions
    - Product Questions
    - Feature Requests
    - General Inquiry
- Auto-Assignment Rules:
    - Technical issues → Technical support staff
    - Payment issues → Finance team
    - Complaints → Dedicated complaint handler
    - High priority → Senior admin
- Acceptance Criteria:
    - Users can select category when creating ticket
    - Auto-assignment routes correctly
    - Can manually reassign if needed

FR-AD-26: Support Analytics

- Priority: Low
- Description: Track support team performance.
- Metrics:
    - Total tickets (by period)
    - Average response time
    - Average resolution time
    - Tickets by category
    - Resolution rate
    - Customer satisfaction rating
- Team Performance:
    - Tickets handled per admin
    - Average response time per admin
    - Resolution rate per admin
- Acceptance Criteria:
    - Metrics calculate accurately
    - Can filter by date range

        - Export reports

5.3.7 Notifications & Messaging

FR-AD-27: Send Platform Announcements

- Priority: High
- Description: Send broadcast messages to users.
- Announcement Creation:
    - Select recipient group:
        - All Users
        - All Customers
        - All Sellers
        - Specific Category Sellers
        - Custom segment
    - Enter subject line
    - Compose message (rich text editor)
    - Select delivery channels:
        - In-app notification
        - Email
        - Both
    - Schedule send (immediate or future date/time)
- Use Cases:
    - Platform updates
    - New features
    - Maintenance notifications
    - Promotional campaigns
    - Policy changes
- Acceptance Criteria:
    - Message delivers to selected recipients
    - Scheduled messages send at correct time
    - Delivery confirmation provided

FR-AD-28: Automated Notification Configuration

- Priority: Medium
- Description: Configure automated system notifications.
- Notification Types:
    - Order confirmations
    - Shipping updates
    - Payment reminders
    - Password resets
    - Account activations
    - Low stock alerts
    - Payout confirmations
- Configuration:
    - Enable/disable specific notifications
    - Customize message templates

        - Set trigger conditions
        - Choose delivery channels
    - Templates:
        - Pre-defined templates with variables
        - Variables: {username}, {orderid}, {amount}, etc.
        - Can edit template content
        - Preview before saving
    - Acceptance Criteria:
        - Templates save correctly
        - Variables populate with actual data
        - Disabled notifications don't send

FR-AD-29: Message Logs

- Priority: Low
- Description: View history of all platform messages sent.
- Display:
    - Log showing:
        - Message subject
        - Recipient(s)
        - Delivery channel
        - Send date/time
        - Status (Sent / Failed / Pending)
- Features:
    - Filter by date, type, recipient group
    - Search by subject or content
    - View delivery statistics
    - Resend failed messages
- Acceptance Criteria:
    - All messages logged
    - Failed deliveries clearly indicated
    - Can investigate delivery issues

FR-AD-30: Email Integration

- Priority: High (if email notifications required)
- Description: Configure email service for platform notifications.
- Configuration:
    - SMTP server settings
    - Sender email address and name
    - Email templates styling
    - Unsubscribe mechanism
- Features:
    - Test email functionality
    - Track email open rates
    - Track click-through rates
    - Manage bounce list
- Acceptance Criteria:

        - Emails deliver successfully
        - Email styling renders correctly across clients
        - Unsubscribe works properly

5.3.8 Platform Settings

FR-AD-31: Category Management

- Priority: High
- Description: Manage business categories and subcategories.
- Current Categories (MVP):
    - Fashion & Accessories
    - Handmade Crafts
    - Natural & Beauty Products
- Actions:
    - Add new category (Phase 2)
    - Edit category name/description
    - Set category icon/image
    - Reorder categories
    - Disable category (hide from users)
- Subcategories:
    - Add subcategories under main categories
    - Assign products to subcategories
- Acceptance Criteria:
    - Category changes reflect immediately on frontend
    - Products reassign correctly
    - Disabled categories hidden but data preserved

FR-AD-32: Commission & Fee Configuration

- Priority: High
- Description: Set platform fees and commission rates (for the platform itself)
- Settings:
    - Platform commission (% per sale)
    - Transaction fee (fixed or %)
    - Minimum payout amount
    - Payout processing fee
- Flexible Settings:
    - Different rates by category (optional)
    - Different rates for featured products (optional)
    - Tiered rates based on seller performance (future)
- Acceptance Criteria:
    - Commission calculates correctly on all transactions
    - Changes apply to new transactions only (not retroactive)
    - Sellers see fee structure transparently

FR-AD-33: Shipping Configuration

- Priority: Medium
- Description: Configure shipping options and costs.

- Settings:
    - Flat rate shipping (per order or per item)
    - Free shipping threshold
    - Shipping zones (domestic, international)
    - Estimated delivery times by zone
- Acceptance Criteria:
    - Shipping costs calculate correctly at checkout
    - Free shipping applies when threshold met
    - Delivery estimates display to customer

FR-AD-34: Tax Configuration

- Priority: Medium (Low for MVP if not required)
- Description: Configure tax settings for the platform(for government)
- Settings:
    - Enable/disable tax collection
    - Tax rate (%)
    - Tax-exempt categories (if applicable)
    - Tax display (inclusive or exclusive)
- Acceptance Criteria:
    - Tax calculates correctly
    - Display matches configuration
    - Compliance with local regulations [SPECIFY]

FR-AD-35: Homepage Management

- Priority: High
- Description: Manage featured content on homepage.
- Manageable Elements:
    - Hero banner (upload image, set link, set text overlay)
    - Featured brands (select up to 10)
    - Featured products (select up to 20)
    - Promotional banners
    - Announcement bar
- Features:
    - Drag-and-drop reordering
    - Schedule features (start and end date)
    - Preview changes before publishing
- Acceptance Criteria:
    - Changes publish immediately or on schedule
    - Preview matches live version
    - Expired features auto-hide

FR-AD-36: Terms & Policies

- Priority: High
- Description: Manage legal and policy documents.
- Documents:
    - Terms of Service
    - Privacy Policy

        - Return & Refund Policy
        - Seller Agreement
        - Community Guidelines
    - Features:
        - Edit document content (rich text editor)
        - Version control (track changes)
        - Publish new version
        - Require user acceptance of new terms
    - Acceptance Criteria:
        - Documents accessible from footer
        - Version history maintained
        - Users notified of policy updates

5.3.9 Vendor Management

FR-AD-37: Vendor List

- Priority: High
- Description: Manage raw material suppliers/vendors.
- Display:
    - Table showing:
        - Vendor Name
        - Material Types
        - Contact Information
        - Rating (optional)
        - Status (Active / Inactive)
        - Actions
- Features:
    - Search by vendor name
    - Filter by material type
    - Sort by name, rating
- Acceptance Criteria:
    - All vendors displayed
    - Search and filters work
    - Only active vendors visible to sellers

FR-AD-38: Add/Edit Vendor

- Priority: High
- Description: Create and modify vendor records.
- Vendor Information:
    - Vendor name
    - Business description
    - Material categories offered
    - Contact person
    - Email
    - Phone
    - Address

        - Website (optional)
    - Associated Materials:
        - Add products (materials) under this vendor
        - Set prices and stock levels
    - Acceptance Criteria:
        - Vendor saves with all information
        - Materials link correctly to vendor
        - Changes reflect in Raw Material Store

FR-AD-39: Vendor Performance Tracking

- Priority: Low
- Description: Monitor vendor reliability and quality.
- Metrics:
    - Total orders from sellers
    - Average order value
    - Fulfillment time
    - Seller ratings/feedback
    - Issue rate (complaints)
- Use Cases:
    - Identify top vendors for featuring
    - Identify problem vendors for removal
- Acceptance Criteria:
    - Metrics calculate from actual orders
    - Can compare vendors
    - Poor performers can be flagged

5.3.10 Educational Content Management

FR-AD-40: Content Library

- Priority: Medium
- Description: Manage educational resources for platform users.
- Display:
    - List of all content items showing:
        - Title
        - Type (Video / PDF / Article)
        - Category (Business / Craft / Marketing / Platform Guide)
        - Status (Published / Draft)
        - Views
        - Last Updated
        - Actions
- Features:
    - Search by title
    - Filter by type, category, status
    - Sort by views, date
- Acceptance Criteria:
    - All content items visible

        - Published content appears in Knowledge Base
        - Draft content not visible to users

FR-AD-41: Upload Content

- Priority: Medium
- Description: Add new educational content.
- Content Types:
    - Video: Upload file or paste YouTube link
    - PDF: Upload document
    - Article: Write in rich text editor
- Metadata:
    - Title
    - Description/summary
    - Category
    - Tags (for searchability)
    - Thumbnail image
    - Target audience (Sellers)
- Acceptance Criteria:
    - Files upload successfully (size limits enforced)
    - YouTube links embed correctly
    - Content categorizes properly

FR-AD-42: Edit/Delete Content

- Priority: Medium
- Description: Modify or remove existing content.
- Edit:
    - Update any metadata
    - Replace file (for videos/PDFs)
    - Edit article content
    - Publish draft or unpublish published content
- Delete:
    - Confirmation required
    - Soft delete (mark as deleted but retain in database)
    - Can restore deleted content
- Acceptance Criteria:
    - Edits save correctly
    - Unpublished content hidden from users
    - Deleted content no longer accessible

FR-AD-43: Content Analytics

- Priority: Low
- Description: Track engagement with educational content.
- Metrics:
    - View count per content item
    - Average view duration (for videos)
    - Most popular content
    - Search terms used

- Use Cases:
    - Identify gaps in educational resources
    - Promote popular content
    - Remove or update unpopular content
- Acceptance Criteria:
    - Metrics track accurately
    - Can sort by popularity
    - Insights actionable

5.3.11 Admin Profile & Access Control

FR-AD-44: Admin Account Management

- Priority: Critical (Super Admin only)
- Description: Create and manage admin user accounts.
- Roles:
    - Super Admin (full access, including admin management)
    - Admin (full platform management, no admin management)
    - Support Staff (limited to tickets and complaints)
    - Content Manager (limited to content and settings)
- Create Admin:
    - Enter name, email
    - Assign role
    - Set temporary password
    - Email credentials to new admin
- Edit Admin:
    - Update information
    - Change role
    - Suspend/reactivate account
    - Reset password
- Acceptance Criteria:
    - Only Super Admin can access this section
    - New admins receive welcome email
    - Role permissions enforce correctly

FR-AD-45: Role-Based Access Control

- Priority: Critical
- Description: Enforce permissions based on admin roles.
- Permission Enforcement:
    - UI hides inaccessible features
    - API validates permissions on every request
    - Unauthorized access attempts logged
- Audit:
    - Can review what each role can access
    - Changes to roles logged
- Acceptance Criteria:
    - Permissions enforce consistently

        - No privilege escalation possible
        - Clear error messages for unauthorized access

FR-AD-46: Activity Logs

- Priority: High
- Description: Track all administrative actions for security and accountability.
- Logged Actions:
    - User account changes (block, delete, etc.)
    - Product approvals/rejections
    - Order interventions
    - Refund processing
    - Setting changes
    - Content modifications
    - Admin logins
- Log Details:
    - Timestamp
    - Admin user
    - Action type
    - Affected entity (user, product, order, etc.)
    - Before/after values (for edits)
    - IP address
- Features:
    - Search logs
    - Filter by action type, admin, date
    - Export logs for auditing
- Acceptance Criteria:
    - All significant actions logged
    - Logs immutable (cannot be deleted/edited)
    - Searchable and exportable

FR-AD-47: Security Settings

- Priority: High
- Description: Configure admin account security.
- Settings:
    - Password complexity requirements
    - Session timeout duration
    - Maximum login attempts before lockout
    - 2FA enforcement for admins
    - IP whitelist (optional)
    - Access time restrictions (optional)
- Acceptance Criteria:
    - Settings apply to all admin accounts
    - 2FA setup is user-friendly
    - Lockout prevents brute force attacks

5.4 AI & Recommendation System

5.4.1 Product Recommendation Engine

FR-AI-01: Customer Product Recommendations

- **Priority:** High
- **Description:** Suggest relevant products to customers based on behavior and preferences.
- **Recommendation Types:**
  - **Personalized Recommendations**: Based on user's browsing history, wishlist, and purchases
  - **Similar Products**: Items similar to currently viewed product
  - **Frequently Bought Together**: Products often purchased with current item
  - **Trending Products**: Popular items across the platform
- **Display Locations:**
  - Homepage: "Recommended for You" section
  - Product detail page: "You May Also Like" and "Customers Also Bought"
  - Cart page: "Complete Your Purchase"
- **Algorithm Approach:**
  - Collaborative filtering (user-based and item-based)
  - Content-based filtering (product attributes)
  - Hybrid approach combining both
- **Data Sources:**
  - User browsing history
  - Purchase history
  - Wishlist items
  - Product categories and tags
  - User demographics
  - Real-time session behavior
- **Acceptance Criteria:**
  - Recommendations appear within 1 second
  - At least 4-6 products suggested when available
  - Recommendations refresh daily or based on new behavior
  - Never recommend out-of-stock items
  - Never recommend items from blocked sellers

FR-AI-02: Cold Start Problem Handling

- **Priority:** Medium
- **Description:** Provide meaningful recommendations for new users without history.

- **Strategies:**
    - Show trending/popular products platform-wide
    - Show top-rated products
    - Show newest arrivals
    - Category-based suggestions (if user indicates interest)
- **Acceptance Criteria:**
    - New users see relevant suggestions immediately
    - Recommendations improve as user interacts
    - System learns user preferences quickly

FR-AI-03: Recommendation Quality Metrics

- **Priority:** Low
- **Description:** Track effectiveness of recommendation engine.
- **Metrics:**
    - Click-through rate on recommendations
    - Conversion rate (recommendation to purchase)
    - Revenue from recommended products
    - User engagement with recommendation sections
- **Use Cases:**
    - Tune recommendation algorithms
    - A/B test different approaches
    - Report to stakeholders on AI impact
- **Acceptance Criteria:**
    - Metrics tracked accurately
    - Dashboard displays performance
    - Can compare different algorithms

5.4.2 Material Recommendation Engine

FR-AI-04: Raw Material Recommendations for Business Owners

- **Priority:** High
- **Description:** Suggest relevant raw materials based on seller's business type and past purchases.
- **Recommendation Logic:**
    - Primary: Based on seller's business category
        - Fashion sellers → Fabrics, threads, buttons, etc.
        - Craft sellers → Resin, wax, dyes, etc.
        - Beauty sellers → Oils, butters, containers, etc.
    - Secondary: Based on past material purchases
    - Tertiary: Based on similar sellers' purchases
    - Quaternary: Trending materials in their category
- **Display Locations:**
    - Raw Material Store homepage: "Recommended for Your Brand"

    - Business Owner Dashboard: "Materials You Might Need"
    - After adding material to cart: "Sellers Also Bought"
- **Seasonal/Trend Awareness:**
    - Promote seasonal materials (e.g., holiday-specific items)
    - Highlight trending materials based on platform-wide demand
- **Acceptance Criteria:**
    - Recommendations relevant to seller's business type
    - At least 6-10 materials suggested
    - Recommendations update based on purchase history
    - Never recommend out-of-stock materials

FR-AI-05: Material Bundling Suggestions

- **Priority:** Low (Phase 2)
- **Description:** Suggest complementary materials often bought together.
- **Examples:**
    - Fabric + matching thread + zippers
    - Resin + hardener + molds
    - Essential oils + carrier oil + bottles
- **Implementation:**
    - Association rule mining (market basket analysis)
    - Display as "Complete Your Supplies" section
- **Acceptance Criteria:**
    - Bundles make logical sense
    - Discount for buying bundle (future enhancement)
    - Easy to add entire bundle to cart

5.4.3 AI Insights Dashboard

FR-AI-06: Sales Trend Analysis

- **Priority:** High
- **Description:** Provide business owners with AI-driven analysis of their sales patterns.
- **Insights Provided:**
    - Sales velocity (trend: increasing, decreasing, stable)
    - Best-selling products identification
    - Peak sales times/days
    - Seasonal patterns
    - Product performance comparison
- **Visualizations:**
    - Line charts for trends over time
    - Bar charts for product comparisons
    - Heatmaps for sales by day/hour
    - Predictive trend lines
- **Actionable Recommendations:**

    - "Your sales spike on weekends - consider running promotions then"
    - "Product X is trending - increase stock"
    - "Sales slowing - try a discount campaign"
- **Acceptance Criteria:**
    - Insights generate from seller's actual data
    - Recommendations are actionable
    - Visualizations are clear and interactive
    - Dashboard loads within 3 seconds

FR-AI-07: Customer Behavior Insights

- **Priority:** Medium
- **Description:** Analyze how customers interact with seller's products.
- **Insights:**
    - View-to-cart conversion rate
    - Cart-to-purchase conversion rate
    - Average time to purchase decision
    - Popular search terms leading to seller's products
    - Cart abandonment rate
    - Customer demographics (age range, location if available)
- **Use Cases:**
    - Optimize product descriptions
    - Identify pricing issues
    - Improve product photography
    - Target marketing efforts
- **Acceptance Criteria:**
    - Insights based on sufficient data (minimum thresholds)
    - Privacy-compliant (anonymized data)
    - Refreshes weekly

FR-AI-08: Competitive Analysis

- **Priority:** Low (Phase 2)
- **Description:** Compare seller's performance to similar sellers in their category.
- **Metrics:**
    - Average price comparison
    - Sales volume comparison
    - Rating comparison
    - Product diversity comparison
- **Presentation:**
    - Anonymized competitor data
    - Percentile ranking
    - Gap analysis with recommendations
- **Acceptance Criteria:**
    - Comparisons are fair (similar category, similar size)

    - Data anonymized to protect competitor privacy
    - Insights help sellers improve

FR-AI-09: Inventory Optimization

- **Priority:** Medium
- **Description:** Recommend optimal stock levels based on sales patterns.
- **Analysis:**
    - Historical sales velocity
    - Seasonal demand patterns
    - Lead time for restocking
    - Storage cost considerations
- **Recommendations:**
    - "Restock Product X - likely to sell out in 5 days"
    - "Reduce stock of Product Y - slow-moving"
    - "Seasonal demand for Product Z increasing - stock up"
- **Acceptance Criteria:**
    - Recommendations consider multiple factors
    - Alerts trigger proactively
    - Accuracy improves over time

5.4.4 Smart Pricing Agent

FR-AI-10: Pricing Recommendations

- **Priority:** High
- **Description:** Suggest optimal pricing for products based on costs, demand, and market conditions.
- **Input Factors:**
    - Material costs (from raw material purchases)
    - Labor/time estimate (seller input)
    - Platform commission and fees
    - Desired profit margin (seller input)
    - Market prices for similar products
    - Product uniqueness/demand
- **Calculation:**
    - Calculate cost base
    - Add desired profit margin
    - Compare to market range
    - Suggest price within competitive range
    - Provide price elasticity estimate
- **Output:**
    - Recommended price
    - Price range (min-max for profitability)
    - Market position (budget/mid-range/premium)

    - Expected demand at suggested price
- **Use Cases:**
    - New product pricing
    - Price adjustments for existing products
    - Promotional pricing suggestions
- **Acceptance Criteria:**
    - Recommendations ensure profitability
    - Market analysis is current
    - Seller can accept or modify suggestion
    - Reasoning for suggestion is transparent

FR-AI-11: Dynamic Pricing Suggestions

- **Priority:** Low (Phase 2)
- **Description:** Suggest price adjustments based on real-time demand.
- **Triggers:**
    - High demand, low stock → suggest price increase
    - Low demand, high stock → suggest discount
    - Seasonal trends → adjust accordingly
    - Competitor pricing changes
- **Implementation:**
    - Notifications to seller with suggestions
    - Optional: Seller pre-authorizes price adjustments within range
- **Acceptance Criteria:**
    - Suggestions are timely
    - Maximize revenue without alienating customers
    - Seller always in control

FR-AI-12: Promotional Pricing Optimization

- **Priority:** Medium
- **Description:** Recommend effective discount strategies.
- **Analysis:**
    - Historical promotion performance
    - Optimal discount percentage for maximum revenue
    - Best timing for promotions
    - Products that benefit most from discounts
- **Recommendations:**
    - "15% discount on Product X could increase sales by 40%"
    - "Bundle Product Y with Product Z for 10% off"
    - "Run promotion on [specific days] for best results"
- **Acceptance Criteria:**
    - Recommendations data-driven
    - ROI projections included
    - Easy to implement suggestions

5.4.5 AI Business Assistant (Chatbot)

FR-AI-13: Conversational Interface

- **Priority:** High
- **Description:** Interactive chatbot to assist business owners with platform navigation and questions.
- **Access Points:**
  - Chat icon in business owner dashboard (bottom right)
  - Quick help button on complex pages
  - Proactive assistance for new users
- **Capabilities:**
  - Answer frequently asked questions
  - Explain dashboard metrics
  - Provide quick tips and best practices
  - Troubleshoot common issues
- **Interaction Flow:**
  - User types question or selects from suggested topics
  - Chatbot provides answer with step-by-step guidance
  - Can escalate to human support if needed
  - Includes helpful links and screenshots
- **Natural Language Processing:**
  - Understands variations of questions
  - Handles typos and informal language
  - Provides relevant responses even with incomplete information
- **Acceptance Criteria:**
  - Responds within 2 seconds
  - Accuracy rate > 80% for common questions
  - Clear escalation path to human support
  - Chat history saved for reference

FR-AI-14: Contextual Help

- **Priority:** Medium
- **Description:** Chatbot provides context-aware assistance based on user's current page/action.
- **Examples:**
  - On "Add Product" page: "Need help creating your first product listing?"
  - After multiple failed login attempts: "Having trouble logging in?"
  - When viewing low sales: "Want tips on improving product visibility?"
- **Proactive Suggestions:**
  - Detects user confusion (rapid page changes, incomplete forms)
  - Offers assistance without being intrusive
  - "It looks like you're trying to [X]. Can I help?"
- **Acceptance Criteria:**

    - Suggestions are timely and relevant
    - Not overly intrusive
    - User can dismiss and not see again

FR-AI-15: Guided Workflows

- **Priority:** Medium
- **Description:** Chatbot walks users through multi-step processes.
- **Workflows:**
    - Adding first product
    - Setting up payment information
    - Processing first order
    - Requesting payout
    - Responding to customer review
- **Implementation:**
    - Step-by-step instructions
    - Visual highlights of relevant UI elements
    - Validation and encouragement
    - Celebration on completion
- **Acceptance Criteria:**
    - Workflows are clear and easy to follow
    - User can pause and resume
    - Completion rate tracked

FR-AI-16: Learning & Improvement

- **Priority:** Low
- **Description:** Chatbot learns from interactions to improve responses.
- **Learning Mechanisms:**
    - Track which responses are helpful (user feedback)
    - Identify questions that escalate to human support
    - Update knowledge base based on common queries
    - A/B test different response styles
- **Admin Tools:**
    - Review unanswered questions
    - Add new Q&A pairs
    - Edit existing responses
    - View chatbot analytics
- **Acceptance Criteria:**
    - Response quality improves over time
    - New patterns identified and addressed
    - Admin can curate chatbot knowledge

5.4.6 Marketing Assistant Agent

FR-AI-17: Marketing Plan Generation

- **Priority:** High
- **Description:** Generate personalized marketing plans for business owners.
- **Input:**
  - Business category
  - Products offered
  - Target audience (seller specifies)
  - Budget (optional)
  - Goals (increase awareness, boost sales, etc.)
- **Generated Plan Includes:**
  - Target audience analysis
  - Recommended platforms (Instagram, Facebook, Pinterest, etc.)
  - Content strategy (post types, frequency)
  - Hashtag suggestions
  - Campaign ideas
  - Timeline (weekly/monthly plan)
  - Budget allocation (if budget provided)
- **Output Format:**
  - Downloadable PDF report
  - Interactive checklist in dashboard
  - Calendar view of scheduled content
- **Acceptance Criteria:**
  - Plan is specific and actionable
  - Recommendations align with business type
  - Budget-conscious suggestions
  - Plan generates within 30 seconds

FR-AI-18: Social Media Content Suggestions

- **Priority:** High
- **Description:** Suggest content ideas and captions for social media posts.
- **Content Types:**
  - Product showcase posts
  - Behind-the-scenes content
  - Customer testimonials
  - Educational/how-to content
  - Promotional posts
  - User-generated content ideas
- **For Each Suggestion:**
  - Post caption (editable)
  - Hashtag recommendations
  - Best time to post
  - Image/video guidance
  - Call-to-action suggestions

- **Personalization:**
    - Based on seller's products
    - Aligned with brand voice (seller can set tone: professional, casual, fun)
    - Seasonal and trending topics
- **Acceptance Criteria:**
    - Suggestions are creative and relevant
    - At least 10 content ideas provided
    - Easy to export or copy
    - Refreshes weekly with new ideas

FR-AI-19: Audience Targeting Recommendations

- **Priority:** Medium
- **Description:** Identify ideal target audience for seller's products.
- **Analysis:**
    - Product category and characteristics
    - Similar successful sellers' audiences
    - Platform demographic data
    - Market trends
- **Recommendations:**
    - Demographic profile (age, gender, location, interests)
    - Where to reach them (platforms, groups, hashtags)
    - Pain points and motivations
    - Messaging angles
- **Use Cases:**
    - Planning ad campaigns
    - Refining product descriptions
    - Choosing marketing channels
- **Acceptance Criteria:**
    - Recommendations are data-driven
    - Specific and actionable
    - Considers seller's niche

FR-AI-20: Campaign Performance Prediction

- **Priority:** Low (Phase 2)
- **Description:** Predict potential reach and engagement for marketing campaigns.
- **Input:**
    - Campaign details (platform, budget, duration)
    - Target audience
    - Content type
- **Output:**
    - Estimated reach
    - Estimated engagement rate
    - Projected conversions

    - ROI estimate
- **Acceptance Criteria:**
    - Predictions based on historical data
    - Accuracy within reasonable range
    - Helps sellers make informed decisions

FR-AI-21: Ad Copy Generator

- **Priority:** Medium
- **Description:** Generate compelling ad copy for product promotions.
- **Input:**
    - Product details
    - Target audience
    - Platform (Facebook, Instagram, Google, etc.)
    - Goal (awareness, conversions, traffic)
- **Output:**
    - Multiple headline options
    - Multiple description options
    - Call-to-action suggestions
    - A/B testing recommendations
- **Style Options:**
    - Professional
    - Friendly
    - Urgent
    - Storytelling
    - Benefit-focused
- **Acceptance Criteria:**
    - Copy is persuasive and relevant
    - Multiple variations provided
    - Adheres to platform character limits
    - Seller can edit before use

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6. Non-Functional Requirements

6.1 Performance Requirements

NFR-01: Page Load Time

- **Requirement:** All pages must load within 3 seconds on a standard broadband connection (5 Mbps).
- **Critical pages** (homepage, product listing, checkout) must load within 2 seconds.

- **Measurement:** 95th percentile of page load times.
- **Acceptance Criteria:** Performance testing validates load times under expected traffic.

NFR-02: API Response Time

- **Requirement:** API endpoints must respond within 500ms for simple queries and 2 seconds for complex queries (reports, analytics).
- **Database queries** should complete within 200ms on average.
- **Acceptance Criteria:** Load testing confirms response times under concurrent users.

NFR-03: Concurrent User Capacity

- **Requirement:** System must support at least 500 concurrent users without performance degradation.
- **Peak capacity:** 1,000 concurrent users during promotional events.
- **Acceptance Criteria:** Load testing simulates concurrent users successfully.

NFR-04: Scalability

- **Requirement:** System architecture must allow horizontal scaling to support growth.
- **Database:** Must handle at least 10,000 products, 5,000 users, and 50,000 orders without performance issues.
- **Acceptance Criteria:** Architecture supports scaling; performance remains acceptable as data grows.

NFR-05: Search Performance

- **Requirement:** Product search must return results within 1 second for queries across entire catalog.
- **Filtering and sorting** must apply instantly (< 500ms).
- **Acceptance Criteria:** Search remains fast with full catalog load.

NFR-06: Real-Time Notifications

- **Requirement:** Real-time notifications (via SignalR) must deliver within 1 second of trigger event.
- **Fallback:** Email notifications sent within 5 minutes if real-time delivery fails.
- **Acceptance Criteria:** Notification delivery time measured and meets requirement.

NFR-07: Image Loading Optimization

- **Requirement:** Product images must be optimized for web delivery.
- **Techniques:** Lazy loading, responsive images, CDN delivery, compression.
- **Acceptance Criteria:** Images load progressively and don't block page rendering.

6.2 Security Requirements

NFR-08: Authentication & Authorization

- **Requirement:** System must implement secure user authentication with password hashing (bcrypt or similar).
- **Authorization:** Role-based access control enforced on every request.
- **Session Management:** Secure session tokens with appropriate expiration.
- **Acceptance Criteria:** Penetration testing validates security; no privilege escalation possible.

NFR-09: Data Encryption

- **Requirement:** All data transmission must use HTTPS/TLS 1.2 or higher.
- **Sensitive data at rest** (passwords, payment info) must be encrypted.
- **Acceptance Criteria:** SSL/TLS certificates properly configured; sensitive data encrypted in database.

NFR-10: Payment Security

- **Requirement:** Payment processing must comply with PCI DSS standards.
- **Implementation:** Use Stripe SDK (PCI-compliant); never store raw card data.
- **Acceptance Criteria:** Security audit confirms PCI compliance; Stripe integration properly implemented.

NFR-11: SQL Injection Prevention

- **Requirement:** All database queries must use parameterized statements or ORMs (Entity Framework).
- **Validation:** Input validation on all user inputs.
- **Acceptance Criteria:** Security testing finds no SQL injection vulnerabilities.

NFR-12: Cross-Site Scripting (XSS) Prevention

- **Requirement:** All user-generated content must be sanitized before display.
- **Implementation:** Output encoding, Content Security Policy headers.
- **Acceptance Criteria:** Security testing finds no XSS vulnerabilities.

NFR-13: Cross-Site Request Forgery (CSRF) Protection

- **Requirement:** All state-changing operations must validate CSRF tokens.
- **Implementation:** Anti-forgery tokens on forms; SameSite cookie attributes.
- **Acceptance Criteria:** CSRF attacks blocked by security measures.

NFR-14: Password Policy

- **Requirement:** Passwords must meet complexity requirements:
    - Minimum 8 characters

    - Mixed case letters
    - At least one number
    - At least one special character (optional but recommended)
- **Storage:** Hashed with bcrypt (work factor ≥ 10).
- **Acceptance Criteria:** Weak passwords rejected; secure storage verified.

NFR-15: Two-Factor Authentication (2FA)

- **Requirement:** 2FA must be available for admin and business owner accounts.
- **Methods:** SMS or authenticator app (TOTP).
- **Acceptance Criteria:** 2FA setup is user-friendly; enforcement works correctly.

NFR-16: Rate Limiting

- **Requirement:** API endpoints must implement rate limiting to prevent abuse.
- **Limits:** [SPECIFY PER ENDPOINT, e.g., 100 requests per minute per user]
- **Acceptance Criteria:** Excessive requests are throttled; legitimate users unaffected.

NFR-17: Audit Logging

- **Requirement:** All sensitive operations must be logged for audit purposes.
- **Logged Actions:** User creation/deletion, permission changes, financial transactions, admin actions.
- **Log Contents:** Timestamp, user, action, affected resources, IP address.
- **Retention:** Logs retained for at least 1 year.
- **Acceptance Criteria:** Comprehensive audit trail; logs immutable.

6.3 Reliability & Availability

NFR-18: System Uptime

- **Requirement:** System must maintain 99.5% uptime (approximately 3.6 hours downtime per month).
- **Planned maintenance:** Scheduled during low-traffic periods with advance notice.
- **Acceptance Criteria:** Uptime monitoring confirms requirement; SLA documented.

NFR-19: Error Handling

- **Requirement:** System must handle errors gracefully without exposing sensitive information.
- **User Experience:** User-friendly error messages; clear recovery instructions.
- **Logging:** All errors logged for debugging.
- **Acceptance Criteria:** No stack traces or sensitive data exposed to users; errors logged correctly.

NFR-20: Data Backup

- **Requirement:** Database must be backed up daily with retention for 30 days.
- **Backup Testing:** Restoration tested monthly.
- **Recovery Time Objective (RTO):** 4 hours.
- **Recovery Point Objective (RPO):** 24 hours (daily backups).
- **Acceptance Criteria:** Backups complete successfully; restoration tested and verified.

NFR-21: Failover & Redundancy

- **Requirement:** Critical components should have redundancy to minimize single points of failure.
- **Database:** Replication or clustering recommended for production.
- **Acceptance Criteria:** System can recover from component failures with minimal downtime.

NFR-22: Transaction Integrity

- **Requirement:** Financial transactions must be atomic (all-or-nothing).
- **Implementation:** Database transactions with proper rollback on failure.
- **Acceptance Criteria:** No partial transactions; data consistency maintained.

6.4 Scalability

NFR-23: Horizontal Scalability

- **Requirement:** Application tier must support horizontal scaling (adding more servers).
- **Stateless Design:** Application servers should be stateless; session data stored centrally.
- **Acceptance Criteria:** Can deploy multiple application instances behind load balancer.

NFR-24: Database Scalability

- **Requirement:** Database design must support growth in users, products, and transactions.
- **Optimization:** Proper indexing, query optimization, partitioning if needed.
- **Acceptance Criteria:** Database performance remains acceptable as data grows 10x.

NFR-25: Asset Storage Scalability

- **Requirement:** Image and file storage must scale independently of application servers.
- **Recommendation:** Use cloud storage (Azure Blob, AWS S3) or CDN.
- **Acceptance Criteria:** Storage solution can handle growing asset library without performance impact.

NFR-26: AI Service Scalability

- **Requirement:** AI microservices must scale independently based on demand.
- **Implementation:** Containerized services (Docker) with orchestration (Kubernetes recommended).
- **Acceptance Criteria:** AI services can handle increased load without affecting core application.

6.5 Usability

NFR-27: Responsive Design

- **Requirement:** User interface must be fully responsive and functional on devices from 320px to 2560px width.
- **Breakpoints:** Mobile (< 768px), Tablet (768-1024px), Desktop (> 1024px).
- **Acceptance Criteria:** UI tested on various devices and screen sizes; all features accessible.

NFR-28: Browser Compatibility

- **Requirement:** Application must function correctly on:
  - Chrome 90+
  - Firefox 88+
  - Safari 14+
  - Edge 90+
- **Acceptance Criteria:** Cross-browser testing validates functionality and appearance.

NFR-29: Accessibility

- **Requirement:** Application should follow WCAG 2.1 Level AA guidelines where feasible.
- **Key Areas:** Keyboard navigation, screen reader compatibility, color contrast, alt text for images.
- **Acceptance Criteria:** Accessibility audit identifies and addresses major issues.

NFR-30: User Onboarding

- **Requirement:** New users should be able to complete key tasks without external help.
- **Implementation:** Tooltips, guided tours, clear CTAs, helpful error messages.
- **Acceptance Criteria:** User testing shows high task completion rate for new users.

NFR-31: Internationalization Readiness

- **Requirement:** Application architecture should support future multi-language expansion.
- **Implementation:** Externalized strings, Unicode support, date/time localization.

- **Acceptance Criteria:** Adding a new language requires minimal code changes (Phase 2 feature).

6.6 Maintainability

NFR-32: Code Quality

- **Requirement:** Code must follow established coding standards and best practices.
- **Standards:** [SPECIFY: C# coding conventions, Angular style guide]
- **Code Reviews:** All code changes must be peer-reviewed.
- **Acceptance Criteria:** Code review process enforced; quality maintained.

NFR-33: Documentation

- **Requirement:** Comprehensive technical documentation must be maintained.
- **Includes:** API documentation, database schema, deployment procedures, architecture diagrams.
- **Acceptance Criteria:** Documentation is up-to-date and accessible to team.

NFR-34: Testing Coverage

- **Requirement:** Automated tests must cover critical functionality.
- **Target:** Minimum 70% code coverage for backend; unit tests for business logic.
- **Types:** Unit tests, integration tests, end-to-end tests.
- **Acceptance Criteria:** Test suite runs successfully; coverage targets met.

NFR-35: Logging & Monitoring

- **Requirement:** System must implement comprehensive logging and monitoring.
- **Logging:** Application logs, error logs, performance logs.
- **Monitoring:** Uptime monitoring, performance monitoring, resource utilization.
- **Alerting:** Automated alerts for critical issues (server down, high error rate).
- **Acceptance Criteria:** Monitoring system detects and alerts on issues proactively.

NFR-36: Deployment Process

- **Requirement:** Deployment must be automated and repeatable.
- **CI/CD:** Continuous Integration and Continuous Deployment pipeline recommended.
- **Environments:** Separate development, staging, and production environments.
- **Rollback:** Ability to quickly rollback to previous version if issues arise.
- **Acceptance Criteria:** Deployment process documented and tested; rollback verified.